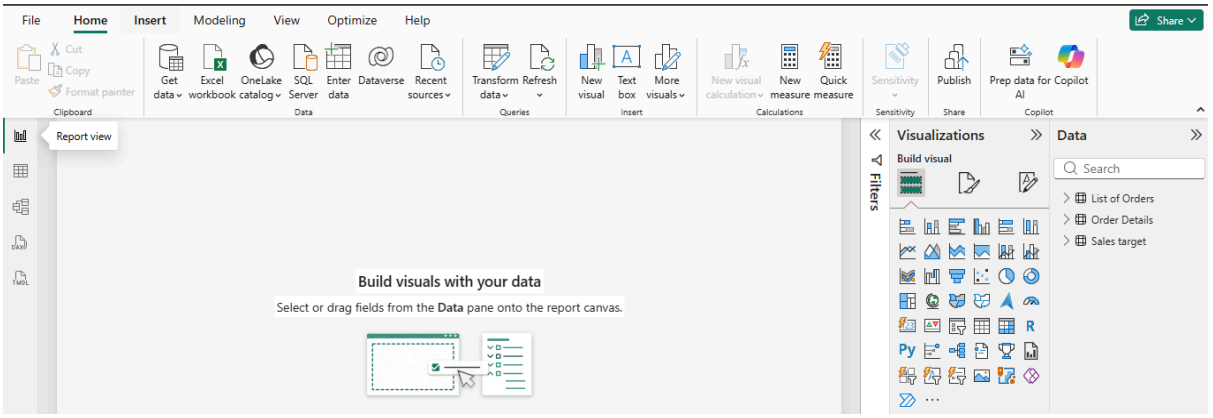
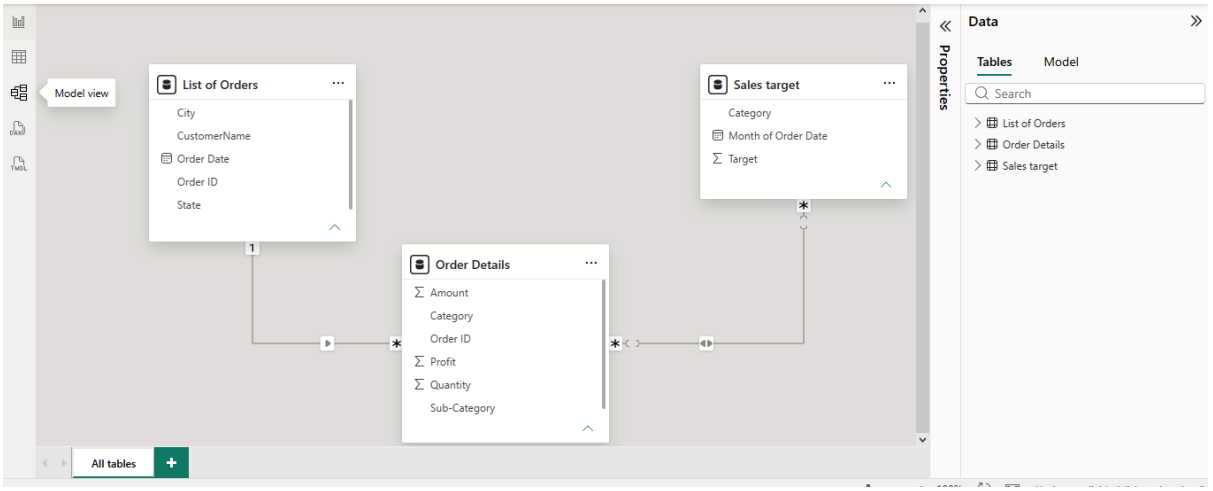
Power BI Assignment 2 – DAX & Data Visualization

E-Commerce Sales Analysis

STEP 1:
Transformed and loaded the given data using Power query

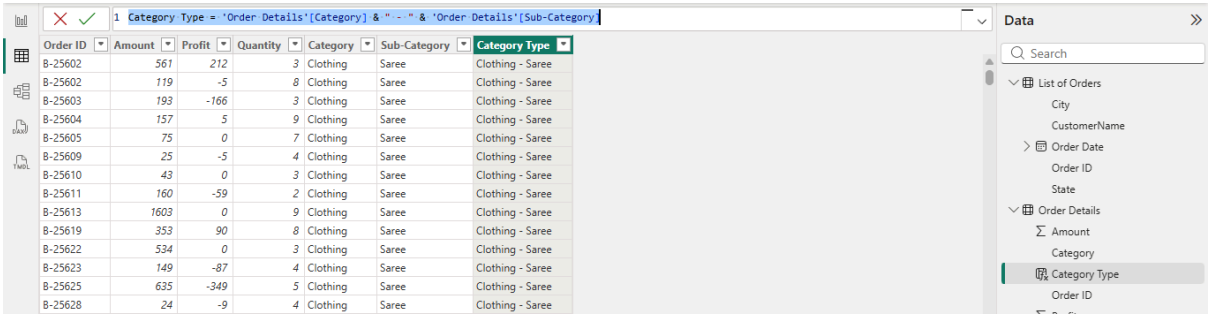


STEP 2:
Relationships between the tables were established



STEP 3:
Calculated Columns:
Added a new column named **"Category Column"** by combining Category and Sub-category column

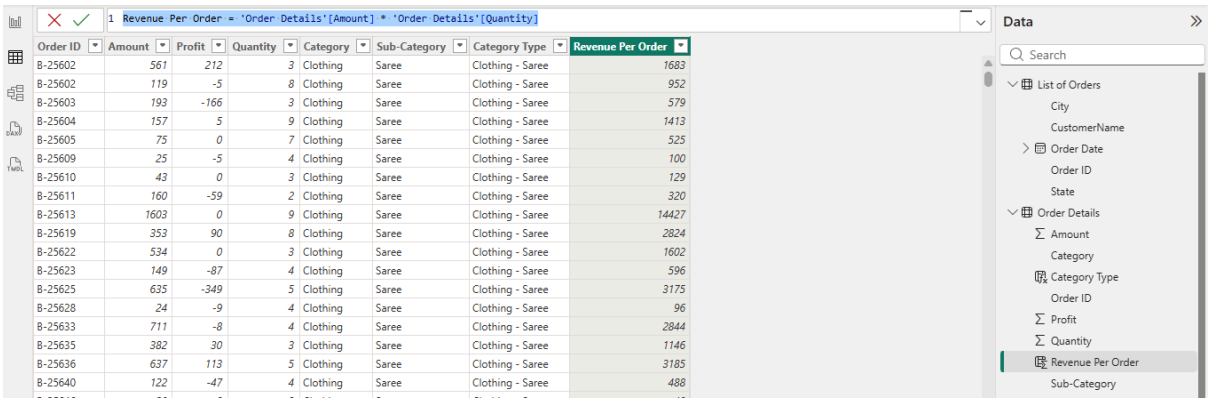
```
Category Type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]
```



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type
B-25602	561	212	3	Clothing	Saree	Clothing - Saree
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree
B-25604	157	5	9	Clothing	Saree	Clothing - Saree
B-25605	75	0	7	Clothing	Saree	Clothing - Saree
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree
B-25610	43	0	3	Clothing	Saree	Clothing - Saree
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree
B-25619	353	90	8	Clothing	Saree	Clothing - Saree
B-25622	534	0	3	Clothing	Saree	Clothing - Saree
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree
B-25628	24	-9	4	Clothing	Saree	Clothing - Saree

Calculated **Revenue Per Order(considering this as sales)** in Order Details table as below

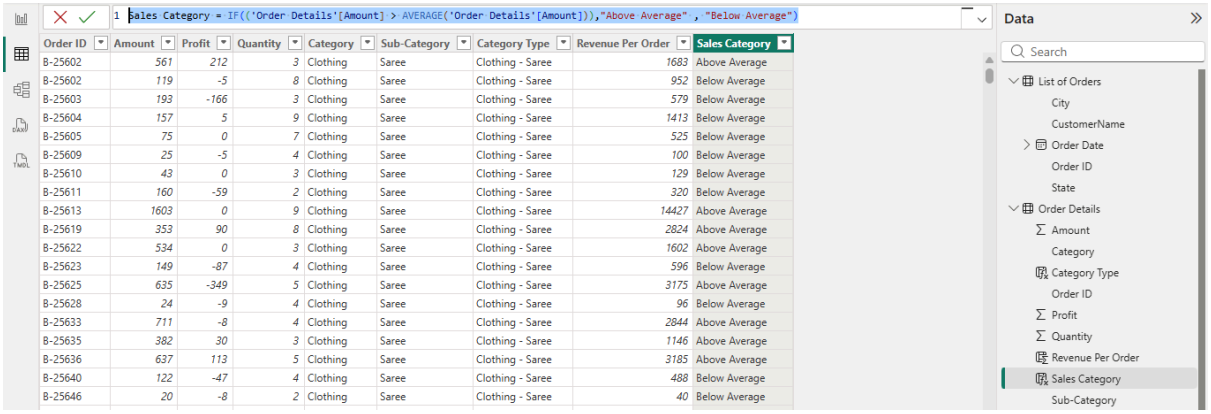
```
Revenue Per Order = 'Order Details'[Amount] * 'Order Details'[Quantity]
```



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue Per Order
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	1683
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	952
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	579
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	1413
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	525
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	100
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	129
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	320
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	14427
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	2824
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	1602
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	596
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	3175
B-25628	24	-9	4	Clothing	Saree	Clothing - Saree	96
B-25633	711	-8	4	Clothing	Saree	Clothing - Saree	2844
B-25635	382	30	3	Clothing	Saree	Clothing - Saree	1146
B-25636	637	113	5	Clothing	Saree	Clothing - Saree	3185
B-25640	122	-47	4	Clothing	Saree	Clothing - Saree	488

Added a calculated column named **‘Sales Category’** in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

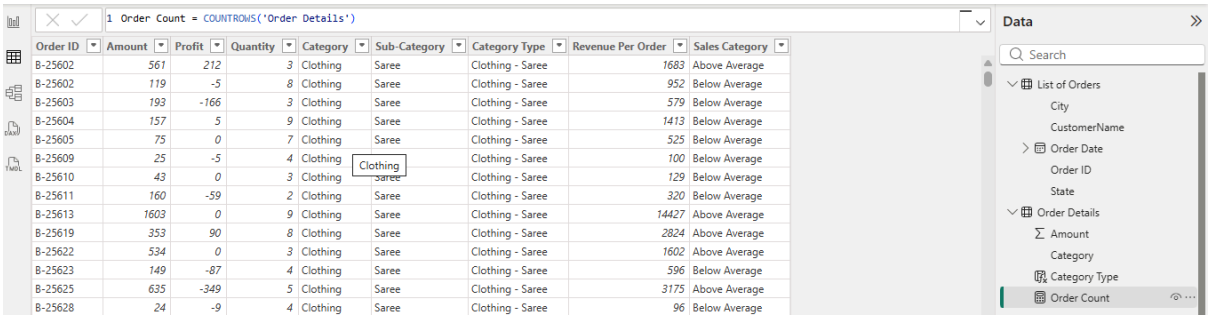
```
Sales Category = IF(('Order Details'[Amount] > AVERAGE('Order
Details'[Amount])),"Above Average" , "Below Average")
```

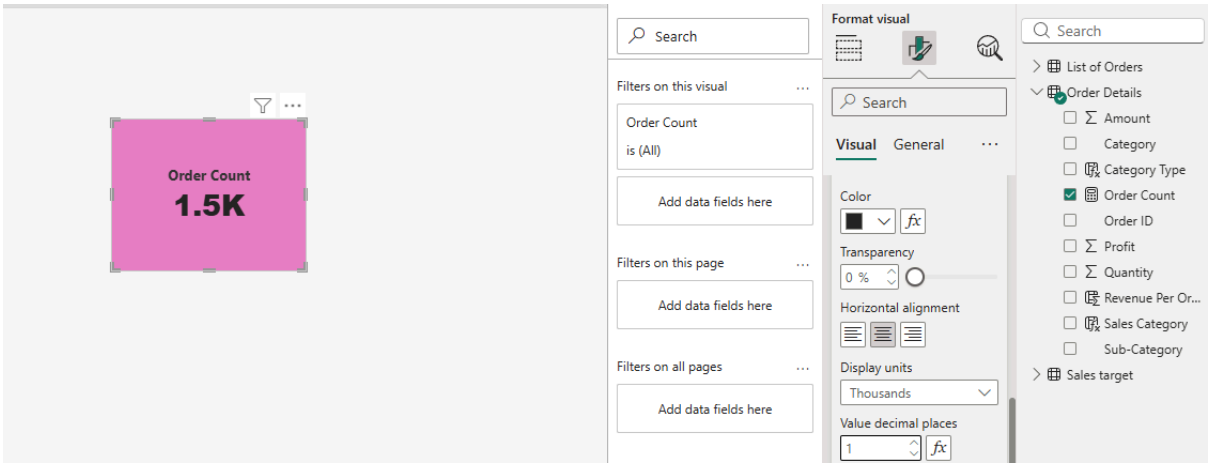


STEP 4:

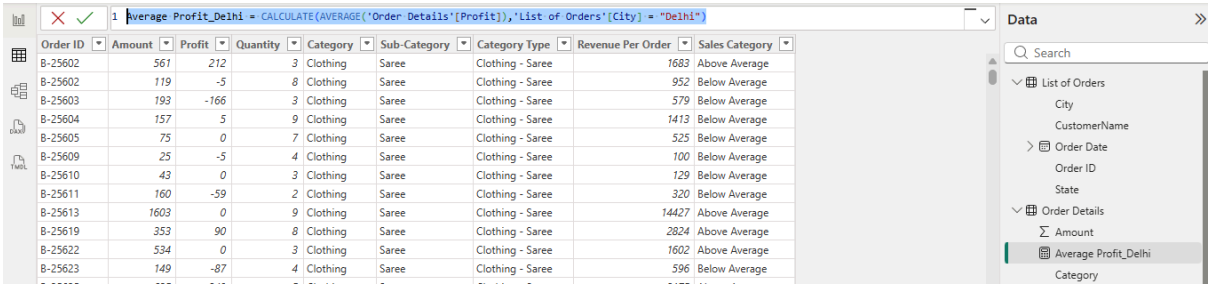
Calculated Measures:

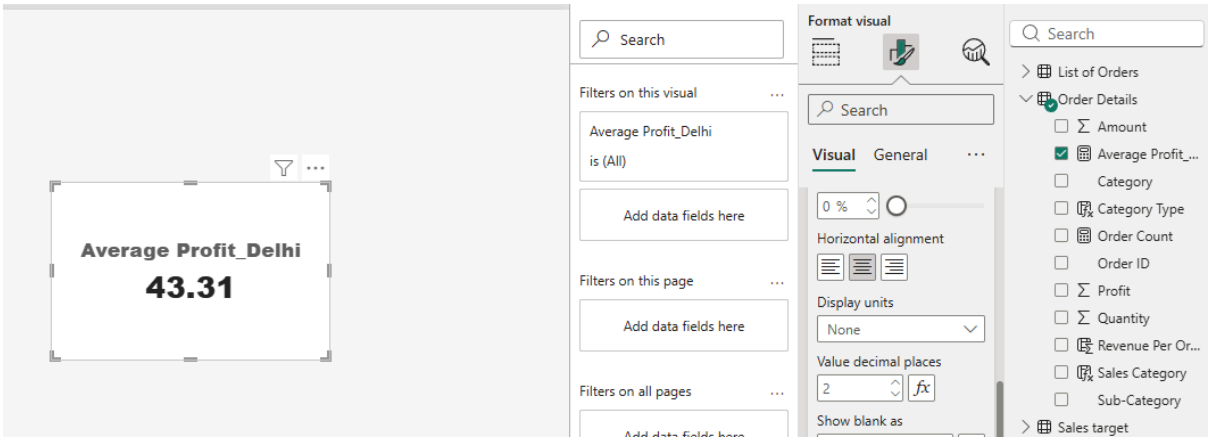
Calculate Order Count: Calculated a measure to count the total number of orders in the Order Details table and displayed the result in a card.



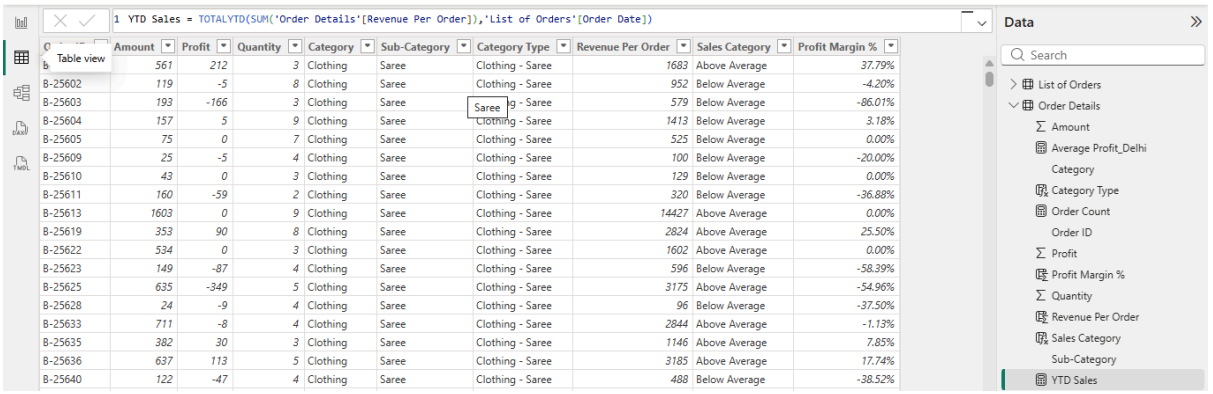


Calculate Average Profit in Delhi: Created a measure to calculate the average profit for orders placed in Delhi.





Calculate Year-to-Date (YTD) Sales: Defined a measure to calculate the total sales amount accumulated from the earliest order date up to each order date.

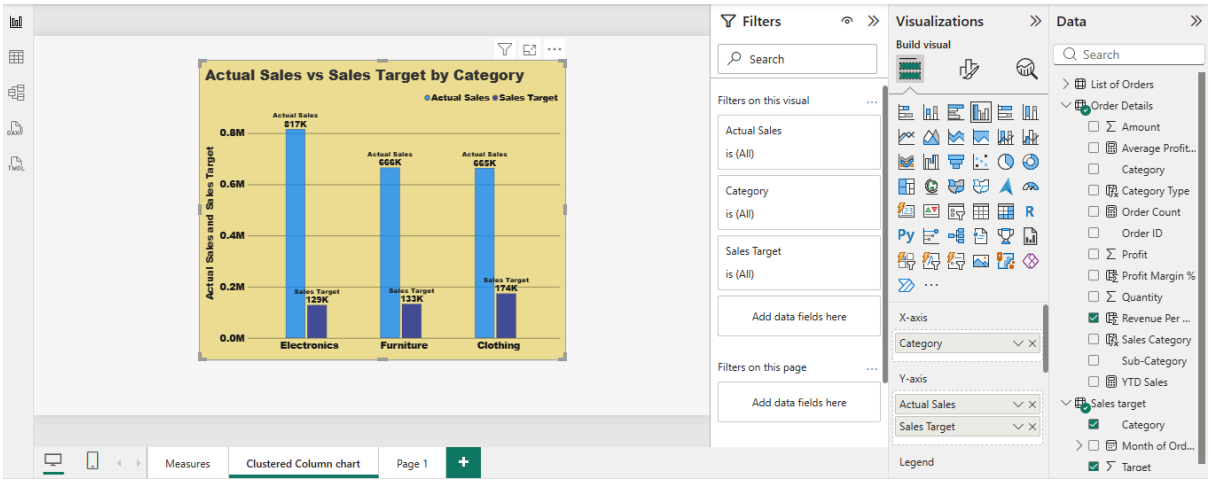


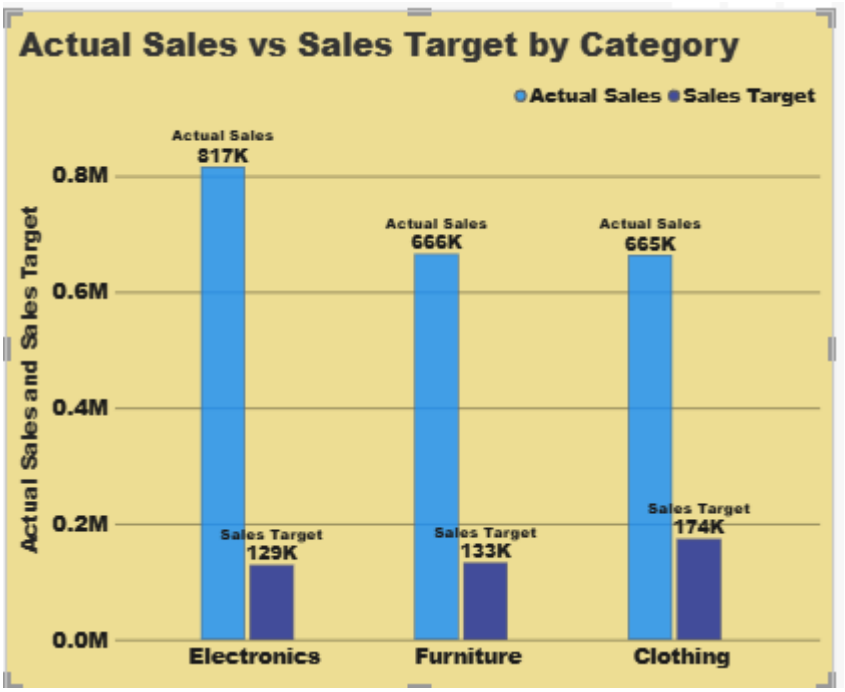


STEP 5:

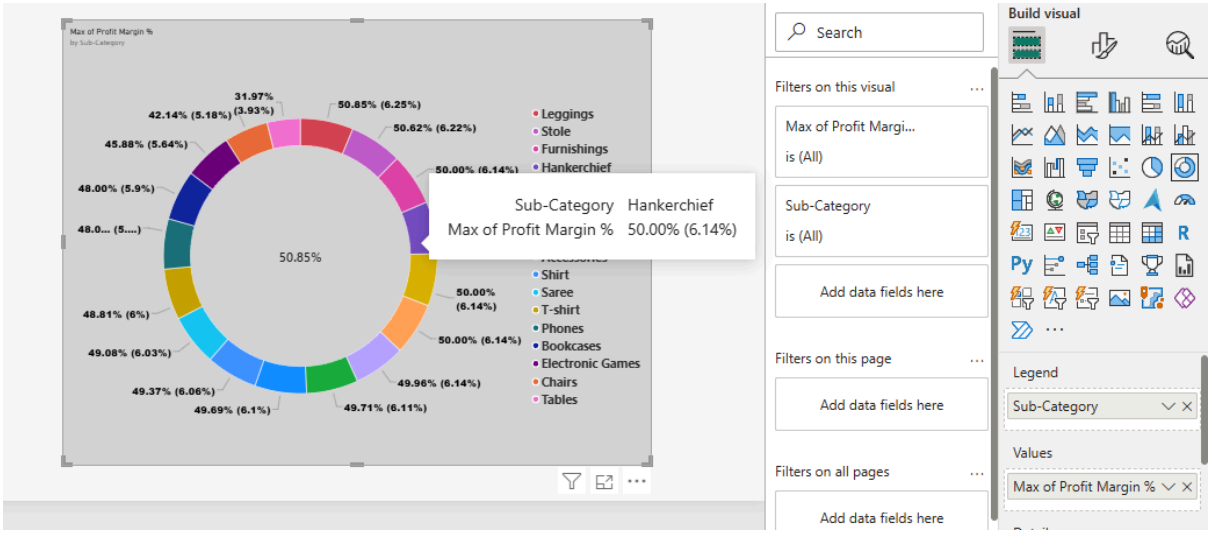
Data Visualization:

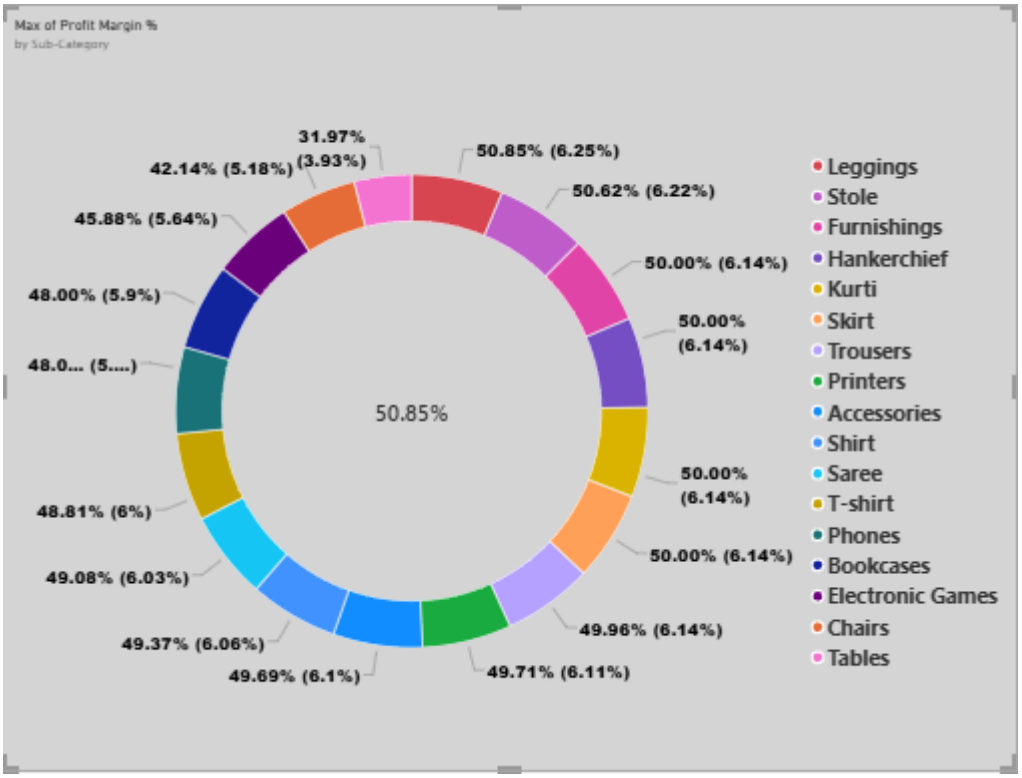
Sales Target Achievement by Category: Compared actual sales with sales targets by category using a clustered column chart.



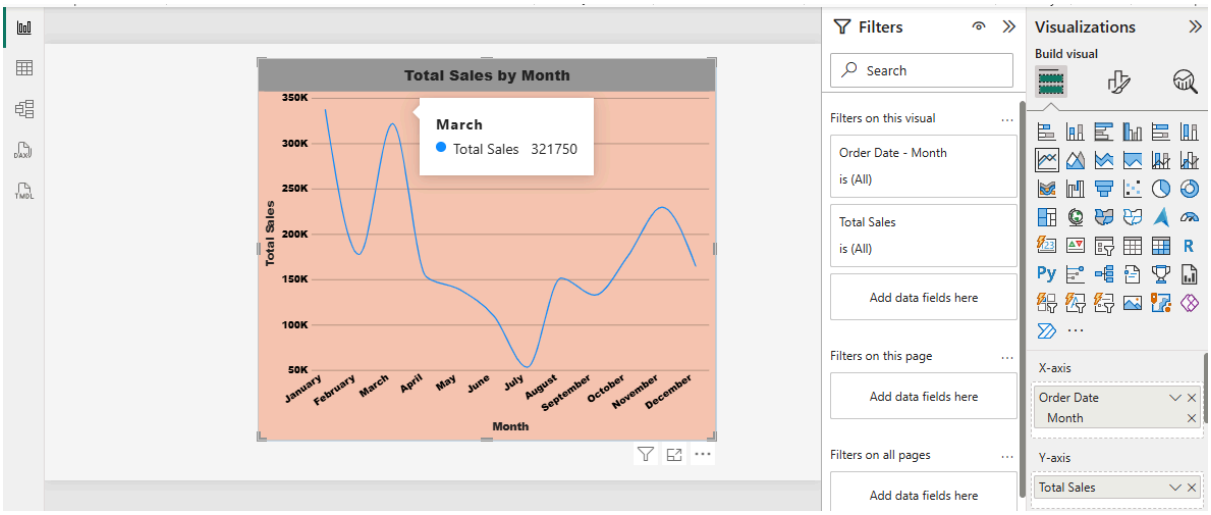


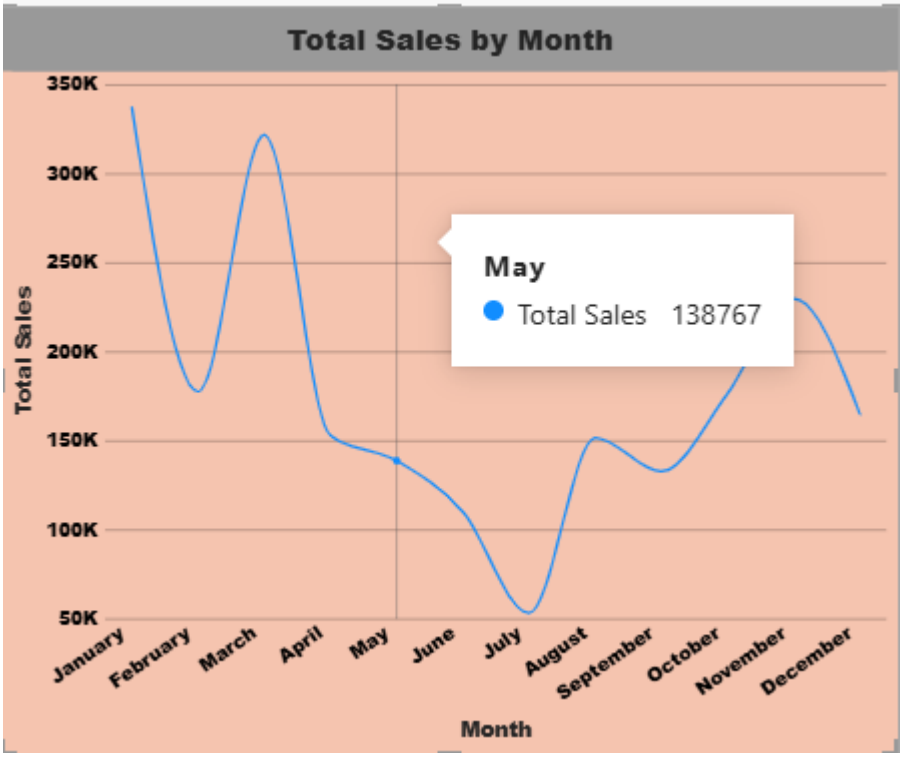
Max Profit Margin by Sub-Category: Calculated a Profit Margin in % column and analyzed the maximum profit margin for each sub-category of products using a donut chart.



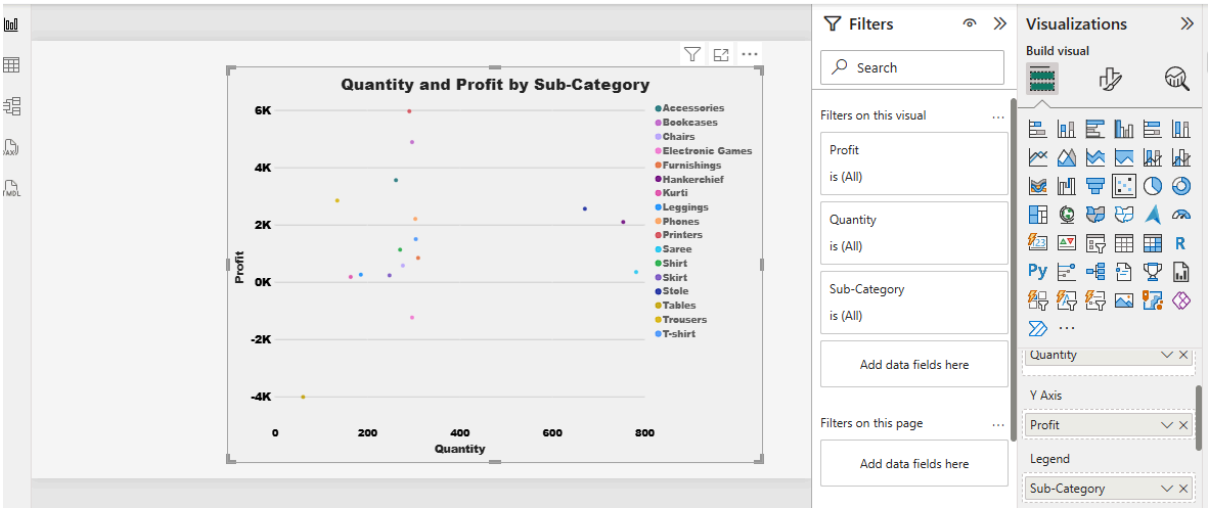


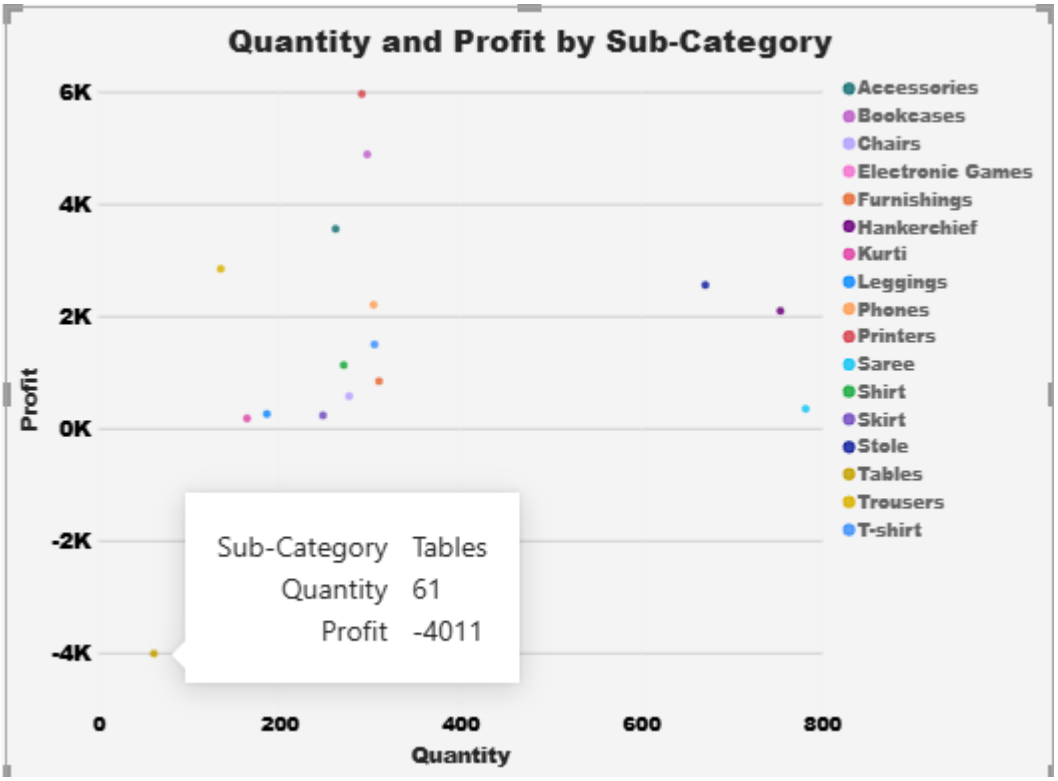
Monthly Sales Trend: Shown the trend of monthly sales over time using a line chart.



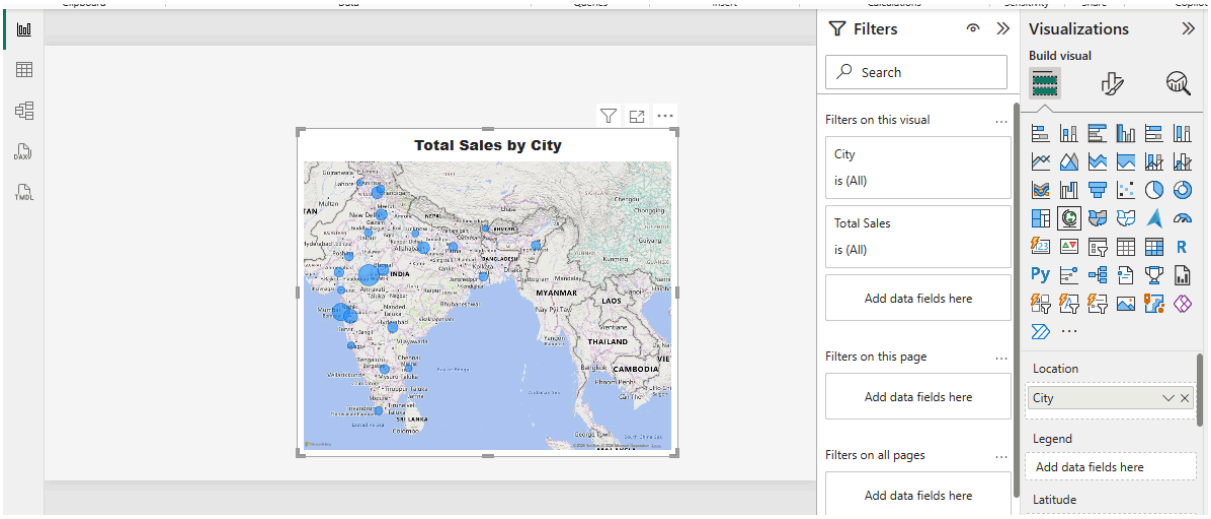


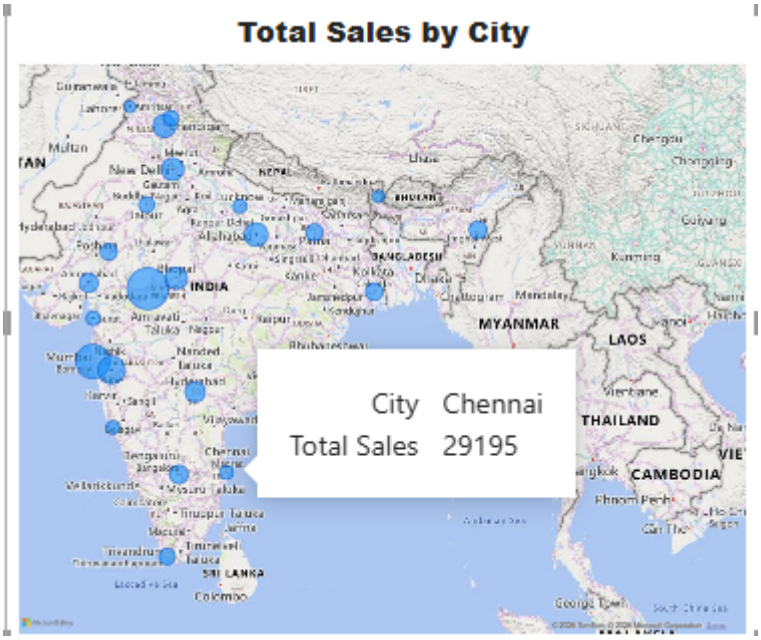
Comparison of Profit and Quantity by Sub-Category: Compared the relationship between profit and quantity sold for different sub-categories using a scatter chart as below



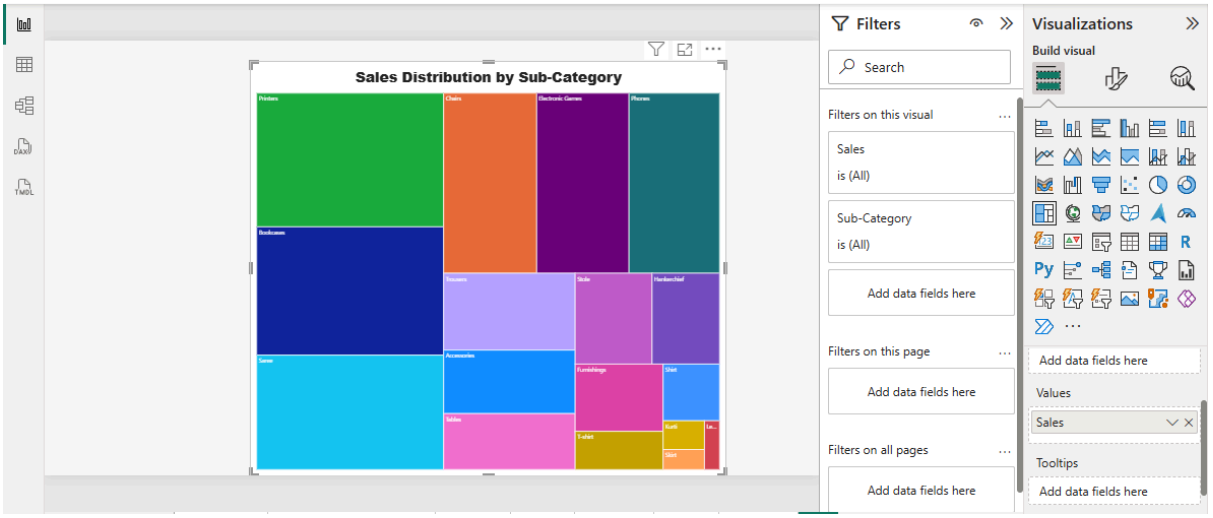


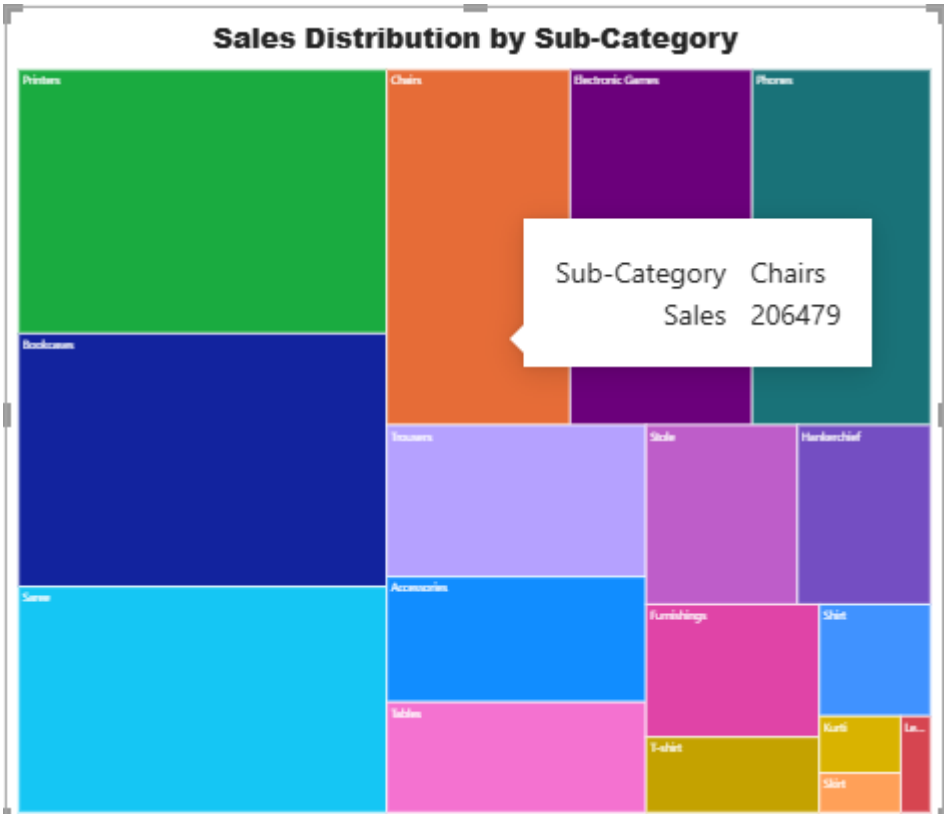
Geographic Sales Analysis: Visualized total sales on a map by city to identify regional sales patterns.



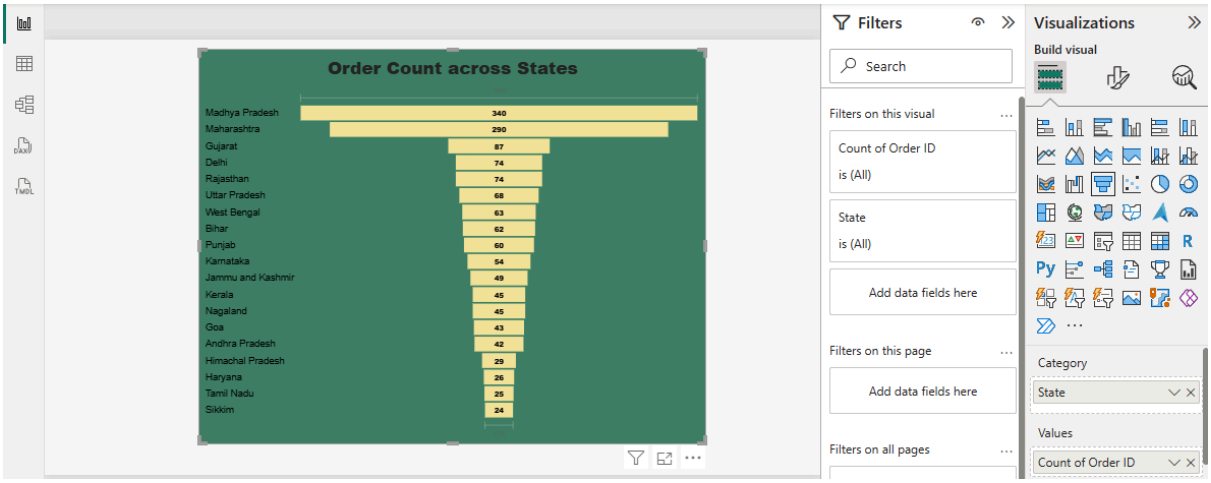


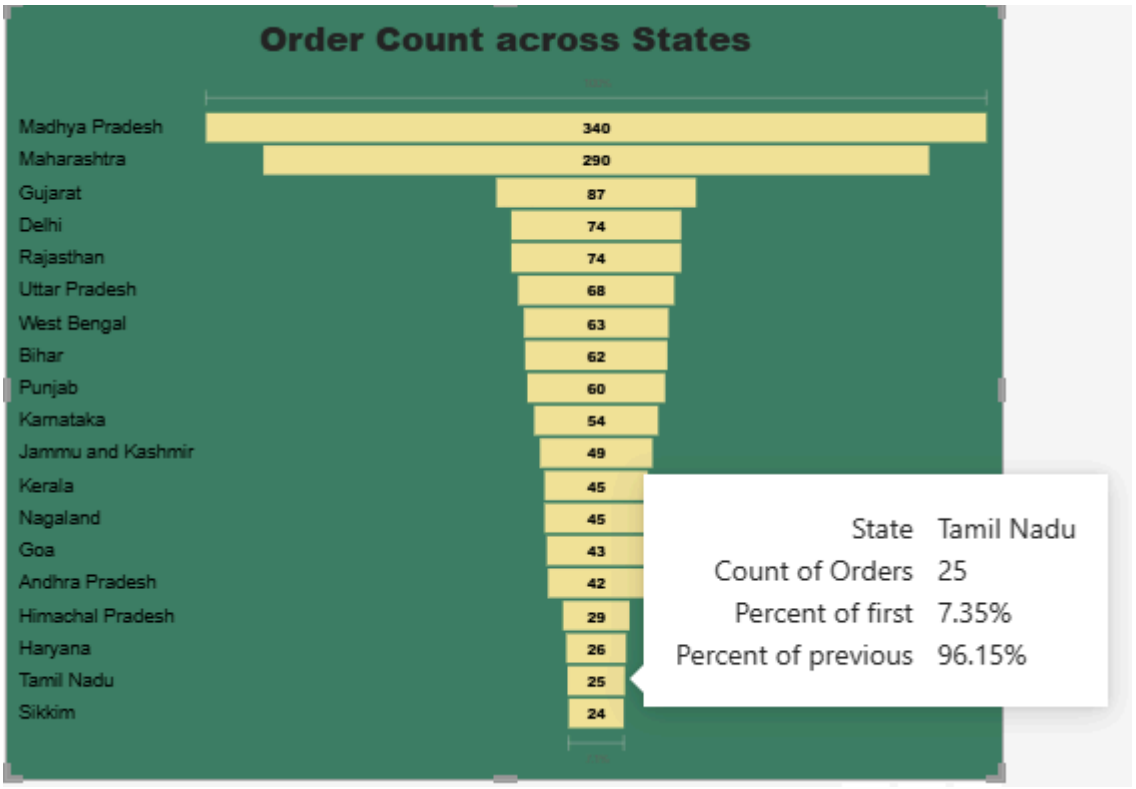
Sales Distribution by Sub-Category: Represented the sales distribution across different sub-categories using a treemap.



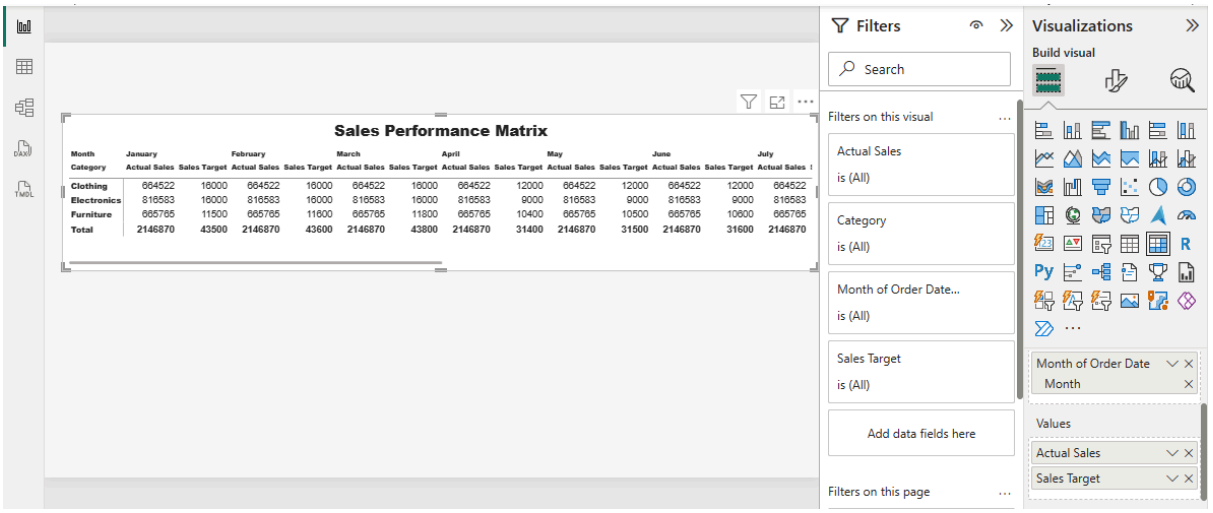


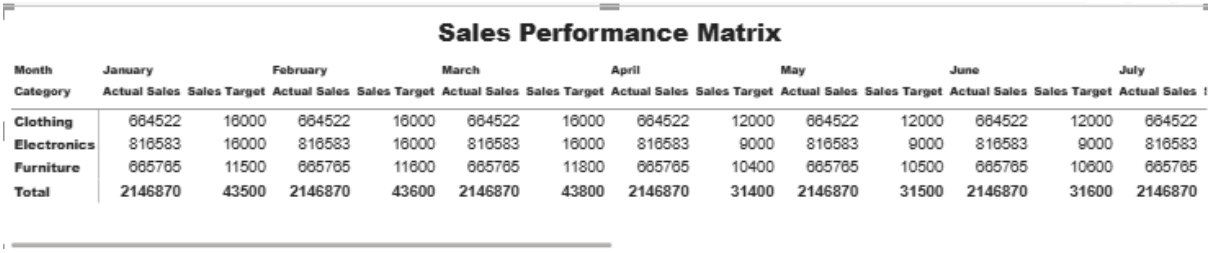
Order Count Analysis by State: Created a funnel chart to visualize the distribution of order counts across different states.





Sales Performance Matrix: Built a matrix view to analyze how actual sales compared to sales targets across different categories and months.

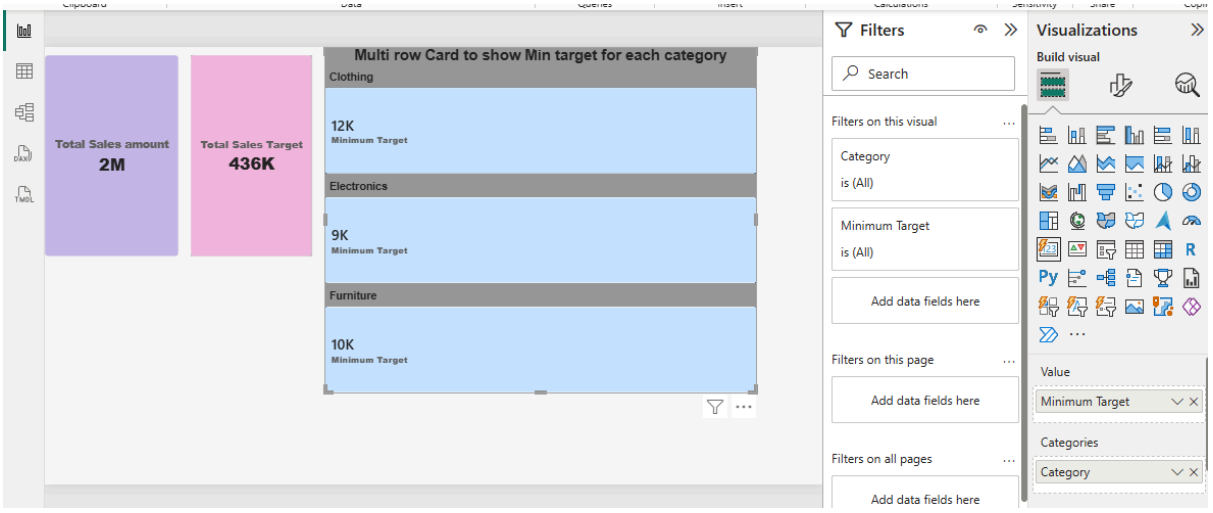


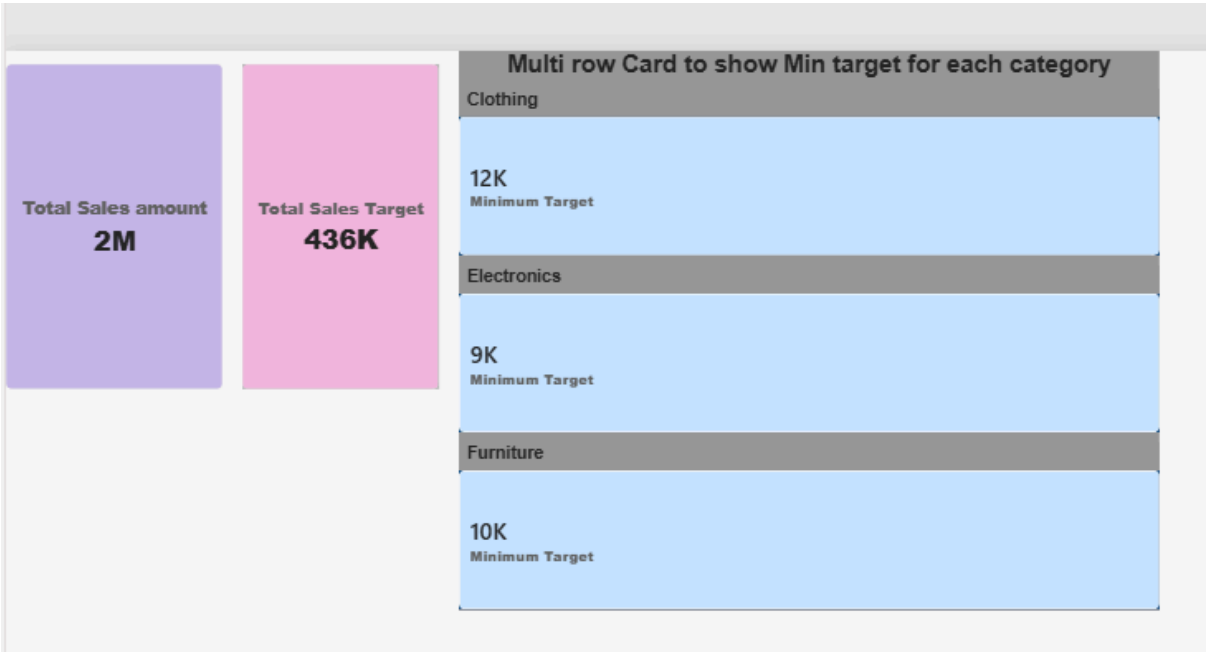


Sales Performance Matrix

Month	January		February		March		April		May		June		July	
Category	Actual Sales	Sales Target	Actual Sales	Sales Target	Actual Sales	Sales Target	Actual Sales	Sales Target	Actual Sales	Sales Target	Actual Sales	Sales Target	Actual Sales	Sales Target	Actual Sales
Clothing	664522	16000	664522	16000	664522	16000	664522	16000	664522	12000	664522	12000	664522	12000	664522
Electronics	816583	16000	816583	16000	816583	16000	816583	16000	816583	9000	816583	9000	816583	9000	816583
Furniture	665765	11500	665765	11600	665765	11800	665765	10400	665765	10500	665765	10600	665765	10600	665765
Total	2146870	43500	2146870	43600	2146870	43800	2146870	31400	2146870	31500	2146870	31600	2146870	31600	2146870

Comparison of Total Sales Amount and Target: Created cards to display the total sales amount alongside the sales target for quick comparison and analysis. Also, created a multi-row card to display the minimum target for each category.




Total Sales amount
2M
Total Sales Target
436K
Multi row Card to show Min target for each category
Clothing
12K
Minimum Target
Electronics
9K
Minimum Target
Furniture
10K
Minimum Target